CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Share of deliveries by autonomous bots (TRA_FR_IL2)

DOMAIN

Transport	Environment	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Innovative last mile solutions** *Increasing the use of autonomous bots for urban logistics*	**TRA_FR**

Category

Key indicator	Supplementary indicator	**State indicator**

CONTEXT AND RELEVANCE

Motorised freight transport refers to the movement of goods using motor vehicles such as trucks and vans. This mode of transport is widely used to deliver goods to customers in urban areas, but it contributes significantly to energy consumption, emissions, noise, and space occupancy. These factors negatively impact environmental sustainability and quality of life in cities.

Alternative solutions for urban deliveries, such as cargo bikes, parcel lockers and drop-off points, autonomous bots, drones, and shared logistics, can help reduce the reliance on conventional motorized freight transport. These alternatives contribute to lower emissions, reduced noise pollution, and improved space efficiency in urban areas.

This indicator provides a measure of the share of deliveries made by autonomous bots in the experiment area. Autonomous bots for urban deliveries are self-navigating robotic systems designed to transport goods without direct human control. These bots typically use a combination of sensors, cameras, and GPS to navigate sidewalks, bike lanes, or roads while avoiding obstacles and ensuring safe interactions with pedestrians and vehicles.

This indicator is relevant when the policy action aims to increase adoption of autonomous bots by logistics providers for delivering goods in the pilot area. A successful action is reflected in a HIGHER value of the indicator.

DESCRIPTION

The indicator is the **share of deliveries made by autonomous bots in the experiment area**.

The indicator is expressed as a **percentage (%)**.

METHOD OF CALCULATION AND INPUTS

The indicator should be computed exogenously, by applying the method described and then coded in the supporting tool.

Method 1

Calculation based on data from logistics operators	Significance: **0.75**

INPUTS

The following information is needed to compute the indicator:

a) The number of deliveries made by autonomous bots by logistics providers in the experiment area over a reference period
b) The total number of deliveries made by logistics providers in the experiment area over the same reference period

The experiment would result in a modification in the number of deliveries by autonomous bots.

METHOD OF CALCULATION

The indicator should be computed exogenously according to the following steps:

- **Retrieval of the number of deliveries by autonomous bots and the total number of deliveries in the experiment area**. If in the 'before' scenario no autonomous bots are used for deliveries in the experiment area, this value equals zero. The data should be obtained from the logistics operators active in the pilot area and should cover a relevant time period, such as an 'average day' reflecting normal operations.

- **Calculation of the index** by computing the ratio between the number of deliveries by autonomous bots and the total number of deliveries in the pilot area.

EQUATIONS

The equation computing the index is the following:

$$\boldsymbol{ABIndex} = \frac{\boldsymbol{ABDel}}{\boldsymbol{TotDel}} \times \mathbf{100}$$

Where:

$ABDel$ = Number of deliveries by autonomous bots in the experiment area over a reference period

$TotDel$ = Total number of deliveries in the experiment area over the same reference period

ALTERNATIVE INDICATORS

This indicator measures the share of deliveries carried out by autonomous bots in the experiment area. To assess the adoption of other innovative last-mile delivery solutions, the framework provides indicator **TRA_FR_IL1** for the share of deliveries made by cargo bikes, **TRA_FR_IL3** for the share of deliveries made by drones, and **TRA_FR_IL4** for the share of deliveries made from microhubs.